Md. Hasibul Islam

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Areas of Research Interest

- **Human-Computer Interaction(HCI) & Human-AI Interaction:** Studying how people understand, interact with and respond to AI-driven systems, with a focus on user agency and trust.
- **Human-Centered AI:** Exploring how AI systems can be designed around human needs, values and expectations to support meaningful user control.
- **Responsible AI:** Investigating fairness, transparency and accountability in AI systems to reduce potential harms and support ethical decision-making.
- **Social Computing:** Understanding how digital platforms and algorithms shape human behavior, social interactions and online communities.

Educational Background

Degree Received: Bachelor of Science in Computer Science & Engineering (CSE)
Instituion: American International University-Bangladesh (AIUB) | Dhaka, Bangladesh | https://www.aiub.edu/
Attendance Dates: 01/2023 - 01/2027
Grade (CGPA): **3.75** out of **4.00**

Professional Credentials

Research Assistant (Remote)
Organization: Elite Research Lab LLC | Queens, NY, USA | https://elitelab.ai/
Description: Conducting research under the guidance of Md. Kishor Morol. A research paper under review at Nature Scientific Reports

Publications

[1] MIME: Multistrategy Imbalance Mitigation Engine for Skin Cancer Classification [link]

- Presented at International Conference on Trends in Computational and Cognitive Engineering(TCCE 2025), Melaka, Malaysia
- Proposed novel approach to handling class imbalance in dermatological AI systems

Under Riview

[2] Reducing Notification Fatigue using a Portable Adaptive-Baseline PM2.5 Monitor for Persistently Polluted Cities (Submitted on 29th ICCIT 2026) [PDF]

- Developed a portable, adaptive PM2.5 monitoring system that detects statistically significant pollution spikes instead of relying on fixed thresholds.
- A two-week field study in Dhaka showed that adaptive spike-based alerts substantially reduced notification frequency and improved the meaningfulness of alerts, helping address notification fatigue.

[3] Testing concept-level audits of melanoma classifiers with diffusion counterfactuals and matched positive controls (Under review at Nature Scientific Reports) [link]

- Develop a diffusion-based generative model to create counterfactual skin-lesion images by controlling seven clinically meaningful dermoscopic concepts.
- Evaluate whether melanoma classifiers respond to these concepts as expected using controlled experiments, matched positive controls, and comparisons with existing explainability methods such as TCAV.

[4] Federated Deep Learning for Alzheimer's Disease Detection: Strong Internal Performance Does Not Survive a Change of Imaging Source (Manuscript ready to submit to Nature Scientific Reports) [PDF]

- A dual-channel ConvNeXt-Small + Swin-V2-S classifier with CBAM fusion, trained centrally and with simulated federated learning (FedAvg, FedProx, FedBN). It reaches 92.74% internal accuracy, but accuracy falls to 12–22% on 86,437 external OASIS-1-derived images. Strong same-source results did not show that the model is robust across imaging sources.

Projects

Undergraduate Academic Projects

Eventique – Event Management System (JavaScript, PHP, HTML, CSS) [GitHub]

- Built a website for an event management system where users can book a hall for their event.
- Signup and signin required before booking
- Admin can maintain(add, delete, update) the whole system.
- Connected the project with a database and followed the **MVC** pattern.

Rural and Urban Scenes – OpenGL project (C++, OpenGL) [GitHub]

- Built a computer graphics project with different scenes using C++ and OpenGL.
- Implemented day-night changes, dynamic object motion, and direction changes.

TaxiCab - Taxi service App Java

- Built a desktop app for a taxi service system.
- Users can book a taxi for a ride; admin can maintain the whole system.

Netspace - ISP Management System .NET, C#

- Built a .NET desktop app for an ISP company.
- Users can book a package and upgrade their current package.
- Admin can add any user, book a package for them, and monitor the whole system.

Volunteering, Leadership & Extracurriculars

- AIUB Research and Development Club – General Member — Dec 2023 – Current
- Bangladesh Scouts – Leadership, teamwork, problem-solving — 2014 – 2018
- BD Clean (General Member) — 2019 – 2021
  volunteer-based youth organization, working to establish Bangladesh as a clean country

Awards & Achievements

- Dean's List Honor - Faculty of Science and Technology
- Academic Scholarship at AIUB: 60% merit-based

Skills

- Programming: C, C++, Python, R, Java, C#
- Data Analysis: Numpy, Pandas, Matplotlib, Seaborn, Plotly
- Machine Learning: TensorFlow, PyTorch
- Tools: Git, LaTeX

Personal References

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